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https://www.tensorflow.org/api_docs/python/tf/clip_by_value

Clips tensor values to a specified min and max.

Function Signatures

```
tf.clip_by_value( t, clip_value_min, clip_value_max,  
name=None )
```

Code Examples

```
tf.clip_by_value(  
t, clip_value_min, clip_value_max, name=None  
)
```

```
tf.clip_by_value(  
t, clip_value_min, clip_value_max, name=None  
)
```

```
t = tf.constant([[ -10., -1., 0.], [0., 2., 10.]])  
t2 = tf.clip_by_value(t, clip_value_min=-1, clip_value_max=1)  
t2.numpy()  
array([[ -1., -1.,  0.],  
       [ 0.,  1.,  1.]], dtype=float32)
```

```
t = tf.constant([[ -10., -1., 0.], [0., 2., 10.]])
```

```
t2 = tf.clip_by_value(t, clip_value_min=-1, clip_value_max=1)
```

```
array([[ -1., -1.,  0.],
```

```
       [ 0.,  1.,  1.]], dtype=float32)
```

Documentation

Clips tensor values to a specified min and max.

View aliases

Compat aliases for migration

See

Migration guide for
more details.

`tf.compat.v1.clip_by_value`

Used in the notebooks

- Matrix approximation with Core APIs
 - Intro to Autoencoders
 - DeepDream
 - Adversarial example using FGSM
 - Neural style transfer
 - Data augmentation

Given a tensor `t`, this operation returns a tensor of the same type and shape as `t` with its values clipped to `clip_value_min` and `clip_value_max`. Any values less than `clip_value_min` are set to `clip_value_min`. Any values greater than `clip_value_max` are set to `clip_value_max`.

For example:

Basic usage passes a scalar as the min and max value.

The min and max can be the same size as `t`, or broadcastable to that size.

Broadcasting fails, intentionally, if you would expand the dimensions of `t`

It throws a `TypeError` if you try to clip an int to a float value
(`tf.cast` the input to float first).

Returns

`## Raises`